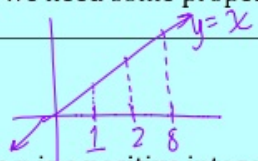
LIM-1.D.1 One-sided limits can be determined analytically or graphically.

LIM-1.D.2 Limits of sums, differences, products, quotients, and composite functions can be found using limit theorems.

1-5 Determining Limits Using Algebraic Properties of Limits

Before we can find limits algebraically, we need some properties:

Basic Properties of Limits	1. $\lim_{x \to c} b = b$ 2. $\lim_{x \to c} x = c$ 3. $\lim_{x \to c} x^n = c^n$, where n is a positive integer.



Other Properties of Limits	Let b and c be real numbers, let n be a positive integer, and let f and g be functions with the following limits: $\lim_{x \to c} f(x) = L$ and $\lim_{x \to c} g(x) = K$ 1. Scalar multiple: $\lim_{x \to c}[bf(x)] = b \cdot \lim_{x \to c}[f(x)] = bL$ 2. Sum or difference: $\lim_{x \to c}[f(x) \pm g(x)] = \lim_{x \to c}[f(x)] \pm \lim_{x \to c}[g(x)] = L \pm K$ 3. Product: $\lim_{x \to c}[f(x) \cdot g(x)] = \lim_{x \to c}[f(x)] \cdot \lim_{x \to c}[g(x)] = L \cdot K$ 4. Quotient: $\lim_{x \to c}\left[\frac{f(x)}{g(x)}\right] = \frac{\lim_{x \to c}[f(x)]}{\lim_{x \to c}[g(x)]} = \frac{L}{K}$ as long as $K \neq 0$ 5. Power: $\lim_{x \to c}[f(x)]^n = \left[\lim_{x \to c}[f(x)]\right]^n = L^n$

Example 1:

Given: $\lim_{x\to 2} f(x) = 4$ $\quad \lim_{x\to 2} g(x) = -2$ $\quad \lim_{x\to 2} h(x) = 0$

Find the following limits. If the limit does not exist, explain why.

(a) $\lim_{x\to 2}[f(x) + 5g(x)] = \lim_{x\to 2} f(x) + \lim_{x\to 2}[5g(x)]$

$= \lim_{x\to 2} f(x) + 5 \cdot \lim_{x\to 2} g(x) = 4 + 5(-2) = -6$

(b) $\lim_{x\to 2} [g(x)]^3 = \left[\lim_{x\to 2} g(x)\right]^3 = (-2)^3 = -8$

(c) $\lim_{x\to 2} \frac{g(x)}{h(x)} = \frac{\lim_{x\to 2} g(x)}{\lim_{x\to 2} h(x)} = \frac{-2}{0}$ DNE b/c $\lim_{x\to 2} h(x) = 0$

(d) $\lim_{x\to 2} \left[\frac{f(x)}{g(x)} + x^3 + \sqrt{f(x)}\right]$

$= \frac{\lim_{x\to 2} f(x)}{\lim_{x\to 2} g(x)} + \lim_{x\to 2}(x^3) + \sqrt{\lim_{x\to 2} f(x)}$

$= \frac{4}{-2} + 8 + \sqrt{4} = -2 + 8 + 2 = \boxed{8}$

Piecewise Functions

Piecewise functions are functions built from pieces of different functions over different intervals. Because of this, they will appear throughout the course in a variety of ways. The laws of limits still apply!

Example 2: The function $f(x)$ is piecewise defined and given below. Use the function to evaluate the following, if possible.

$$f(x) = \begin{cases} 2x - 3, & x < 0 \\ x^2 + e^x, & x \geq 0 \end{cases}$$

equal to

left 0 right

a) $\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} (2x-3) = 0-3 = -3$

b) $\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} (x^2+e^x) = 0^2+e^0 = 0+1 = 1$

c) $\lim_{x \to 0} f(x) =$ DNE b/c $\lim_{x \to 0^-} f(x) \neq \lim_{x \to 0^+} f(x)$

d) $f(0) = 0^2+e^0 = 1$

Example 3: The function $g(x)$ is piecewise defined and given below. Use the function to evaluate the following, if possible.

$$g(x) = \begin{cases} 4 - x^2, & x < 1 \\ 5, & x = 1 \\ 5x - 2, & x > 1 \end{cases}$$

$\rightarrow 1$

a) $\lim_{x \to 1} g(x) = 3$

$\lim_{x \to 1^-} g(x) = \lim_{x \to 1^-} (4 - x^2) = 4 - 1 = 3$

$\lim_{x \to 1^+} g(x) = \lim_{x \to 1^+} (5x - 2) = 5 - 2 = 3$

b) $g(1) = 5$

Example 4: Let $k(x)$ be the piecewise defined function above where c is a constant. Find the value of c such that $\lim_{x\to 2} k(x)$ exists.

$$k(x) = \begin{cases} x^2 - 1, & x \le 0 \\ 7 - cx, & 0 < x < 2 \\ (x+1)^2 - 5, & x > 2 \end{cases}$$

$$\lim_{x\to 2^-} k(x) = \lim_{x\to 2^-} (7 - cx) = 7 - 2c$$

$$\lim_{x\to 2^+} k(x) = \lim_{x\to 2^+} \left[(x+1)^2 - 5\right] = 9 - 5 = 4$$

$$7 - 2c = 4$$

$$-2c = -3$$

$$c = \frac{3}{2}$$

Operations on Limits and Limits of Composite Functions

The standard rule for finding the limit of a composite function looks something like this:

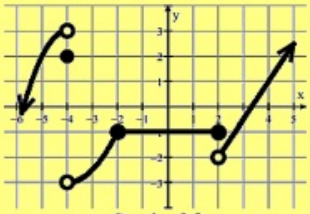
Rule for Limits of Composite Functions

If $\lim_{x \to a} g(x) = K$ and $f(x)$ is continuous at $x = K$, then

$$\lim_{x \to a} f(g(x)) = f\left(\lim_{x \to a} g(x)\right).$$

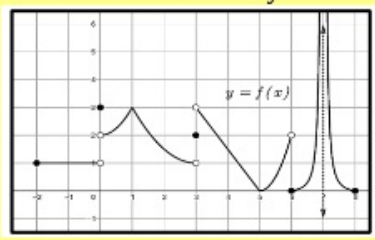
We will expand upon the idea of continuity later in the unit. For now, think of $f(x)$ is continuous at $x = K$ if $f(x)$ has no **holes**, **breaks** or **asymptotes** at $x = K$. While useful, this rule can be very problematic if the inner function's limit does not exist or if the outer function is not continuous at the value of the inner function's limit.

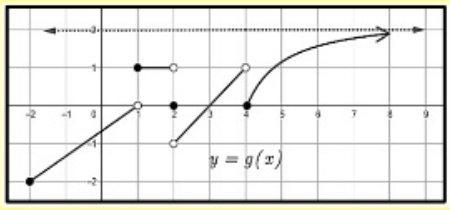
Example 5: The graph of the function f is shown at the right. What is $\lim_{x \to 4} f(f(x))$?



Graph of f

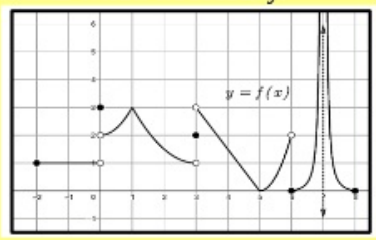
Example 6: Consider the graphs of the functions $f(x)$ and $g(x)$ given below. Find each of the following limits using a pair of one-sided limits when necessary.

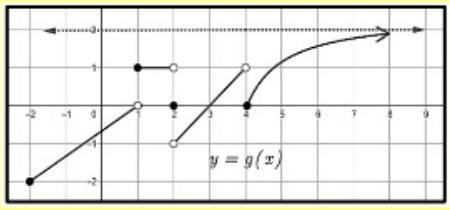




a.) $\lim\limits_{x \to 3} [f(x)g(x)]$

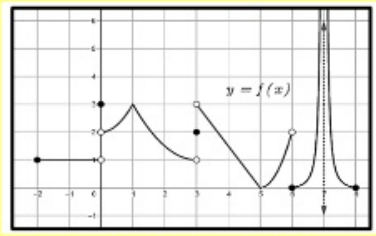
Example 6: Consider the graphs of the functions $f(x)$ and $g(x)$ given below. Find each of the following limits using a pair of one-sided limits when necessary.

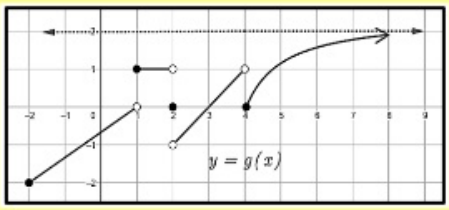




b.) $\lim_{x \to 3} [f(x) + f(x+3)]$

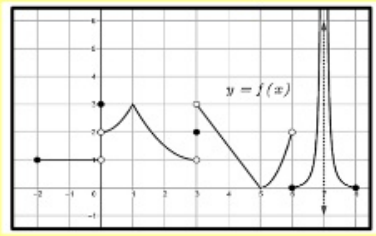
Example 6: Consider the graphs of the functions $f(x)$ and $g(x)$ given below. Find each of the following limits using a pair of one-sided limits when necessary.

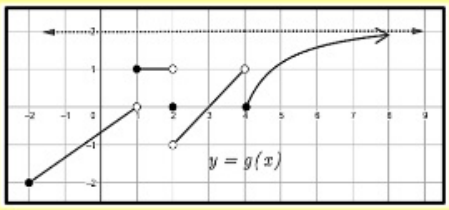




c.) $\lim\limits_{x \to 6}[f(x)f(x-1)]$

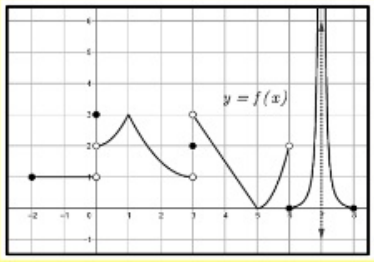
Example 6: Consider the graphs of the functions $f(x)$ and $g(x)$ given below. Find each of the following limits using a pair of one-sided limits when necessary.

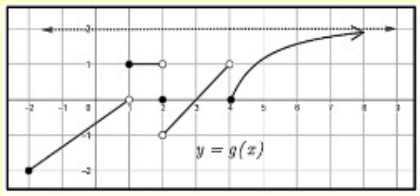




d.) $\lim_{x \to 3} [f(x) + g(x+1)]$

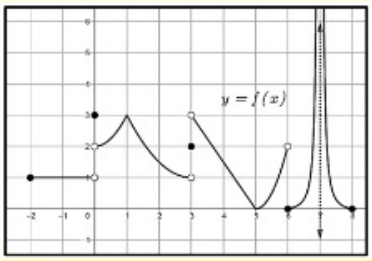
Example 7: Consider the graphs of the functions $f(x)$ and $g(x)$ given below. Find each of the following limits.

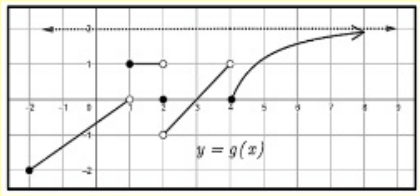




a.) $\lim_{x \to 1} f(f(x))$

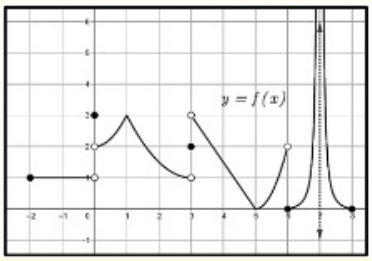
Example 7: Consider the graphs of the functions $f(x)$ and $g(x)$ given below. Find each of the following limits.

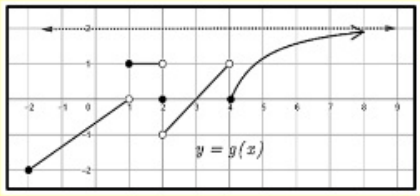




b.) $\lim_{x \to 3} f(g(x))$

Example 7: Consider the graphs of the functions $f(x)$ and $g(x)$ given below. Find each of the following limits.





c.) $\lim_{x \to 7} g(f(x))$